__**Q. What you intend to teach?**__

**1. Defining AI and its purpose: **

Artifcial intelligence (AI) is a foundational technology that is supercharging other scientifc fields and, like electricity and the internet, has the potential to transform societies, economies, and politics worldwide.
(Source: https://setr.stanford.edu/technology/artificial-intelligence/2026)
(Authors: CO-CHAIRS Condoleezza Rice, Jennifer Widom, and Amy Zegart DIRECTOR AND EDITOR IN CHIEF Herbert S. Lin | MANAGING EDITOR Martin Giles)

2. How does AI work?

Below, we'll go through five steps that illustrate how AI works: inputs, processing, outcomes, adjustments, and assessments.

Input
Data is first collected from various sources in the form of text, audio, video, and more. It is sorted into categories, such as those that can be read by the algorithms and those that cannot. You would then create the protocol and criteria for which data will be processed and used for specific outcomes.

Processing
Once data is gathered and input, the next step is to allow AI to decide what to do with the data. The AI sorts and deciphers the data using patterns it has been programmed to learn until it recognizes similar patterns in the data that is being filtered into the system.

Outcomes
After the processing step, the AI can use those complex patterns to predict outcomes in customer behavior and market trends. In this step, the AI is programmed to decide whether specific data is a "pass" or "fail"—in other words, does it match previous patterns? That determines outcomes that can be used to make decisions.

Adjustments
When data sets are considered a "fail," AI learns from that mistake, and the process is repeated again under different conditions. It may be that the algorithm's rules must be adjusted to suit the data set in question or that the algorithm needs slight alteration. In this step, you might return to the outcomes step to better align with the current data set's conditions.

Assessments

The final step for AI to complete an assigned task is assessment. Here, the AI technology synthesizes insights gained from the data set to make predictions based on the outcomes and adjustments. Feedback generated from the adjustments can be incorporated into the algorithm before moving forward. (Coursera Staff, 2026)
(Source: https://www.coursera.org/articles/how-does-ai-work)

****3. Different types of AI?****

__The three kinds of AI based on capabilities__
1. Artificial Narrow AI

Artificial Narrow Intelligence, also known as Weak AI (what we refer to as Narrow AI), is the only type of AI that exists today. Any other form of AI is theoretical. It can be trained to perform a single or narrow task, often far faster and better than a human mind can.

However, it can't perform outside of its defined task. Instead, it targets a single subset of cognitive abilities and advances in that spectrum. Siri, Amazon's Alexa and IBM Watson® are examples of Narrow AI. Even OpenAI's ChatGPT is considered a form of Narrow AI because it's limited to the single task of text-based chat.

2. General AI

Artificial General Intelligence (AGI), also known as Strong AI, is today nothing more than a theoretical concept. AGI can use previous learnings and skills to accomplish new tasks in a different context without the need for human beings to train the underlying models. This ability allows AGI to learn and perform any intellectual task that a human being can.

3. Super AI

Super AI is commonly referred to as artificial superintelligence and, like AGI, is strictly theoretical. If ever realized, Super AI would think, reason, learn, make judgements and possess cognitive abilities that surpass those of human beings.

The applications possessing Super AI capabilities will have evolved beyond the point of understanding human sentiments and experiences to feel emotions, have needs and possess beliefs and desires of their own.

(IBM Data and AI Team, 2024)
(Source: https://www.ibm.com/think/topics/artificial-intelligence-types)

4. Additional capabilities and practical applications of AI technologies
__Computer vision__

Narrow AI applications with computer vision can be trained to interpret and analyze the visual world. This allows intelligent machines to identify and classify objects within images and video footage.

__Applications of computer vision include:__

Image recognition and classification
Object detection
Object tracking
Facial recognition
Content-based image retrieval
Computer vision is critical for use cases that involve AI machines interacting and traversing the physical world around them. Examples include self-driving cars and machines navigating warehouses and other environments.

__Robotics__

Robots in industrial settings can use Narrow AI to perform routine, repetitive tasks that involve materials handling, assembly and quality inspections. In healthcare, robots equipped with Narrow AI can assist surgeons in monitoring vitals and detecting potential issues during procedures.

Agricultural machines can engage in autonomous pruning, moving, thinning, seeding and spraying. And smart home devices such as the iRobot Roomba can navigate a home's interior using computer vision and use data stored in memory to understand its progress.

__Expert systems__

Expert systems equipped with Narrow AI capabilities can be trained on a corpus to emulate the human decision-making process and apply expertise to solve complex problems. These systems can evaluate vast amounts of data to uncover trends and patterns to make decisions. They can also help businesses predict future events and understand why past events occurred.

(IBM Data and AI Team, 2024)
(Source: https://www.ibm.com/think/topics/artificial-intelligence-types)

5. Benefits and Disadvantages of AI

The real-world potential of AI is immense. Applications of AI include diagnosing diseases, personalizing social media feeds, executing sophisticated data analyses for weather modeling and powering the chatbots that handle our customer support requests. AI-powered robots can even assemble cars and minimize radiation from wildfires.

As with any technology, there are advantages and disadvantages of AI, when compared to traditional programing technologies. Aside from foundational differences in how they function, AI and traditional programming also differ significantly in terms of programmer control, data handling, scalability and availability.

__Control and transparency__: Traditional programming offers developers full control over the logic and behavior of software, allowing for precise customization and predictable, consistent outcomes. And if a program doesn't behave as expected, developers can trace back through the codebase to identify and correct the issue. AI systems, particularly complex models like deep neural networks, can be hard to control and interpret. They often work like "black boxes," where the input and output are known, but the process the model uses to get from one to the other is unclear. This lack of transparency can be problematic in industries that prioritize process and decision-making explainability (like healthcare and finance).

__Learning and data handling__: Traditional programming is rigid; it relies on structured data to execute programs and typically struggles to process unstructured data. In order to "teach" a program new information, the programmer must manually add new data or adjust processes. Traditionally coded programs also struggle with independent iteration. In other words, they may not be able to accommodate unforeseen scenarios without explicit programming for those cases. Because AI systems learn from vast amounts of data, they're better suited for processing unstructured data like images, videos and natural language text. AI systems can also learn continually from new data and experiences (as in machine learning), allowing them to improve their performance over time and making them especially useful in dynamic environments where the best possible solution can evolve over time.

__Stability and scalability__: Traditional programming is stable. Once a program is written and debugged, it will perform operations the exact same way, every single time. However, the stability of rules-based programs comes at the expense of scalability. Because traditional programs can only learn through explicit programming interventions, they require programmers to write code at scale in order to scale up operations. This process can prove unmanageable, if not impossible, for many organizations. AI programs offer more scalability than traditional programs but with less stability. The

automation and continuous learning features of AI-based programs enable developers to scale processes quickly and with relative ease, representing one of the key advantages of ai. However, the improvisational nature of AI systems means that programs may not always provide consistent, appropriate responses.

__Efficiency and availability__: Rules-based computer programs can provide 24/7 availability, but sometimes only if they have human workers to operate them around the clock.

AI technologies can run 24/7 without human intervention so that business operations can run continuously. Another of the benefits of artificial intelligence is that AI systems can automate boring or repetitive jobs (like data entry), freeing up employees' bandwidth for higher-value work tasks and lowering the company's payroll costs. It's worth mentioning, however, that automation can have significant job loss implications for the workforce. For instance, some companies have transitioned to using digital assistants to triage employee reports, instead of delegating such tasks to a human resources department. Organizations will need to find ways to incorporate their existing workforce into new workflows enabled by productivity gains from the incorporation of AI into operations.

(Chrystal R. China, 2024)
(Source: https://www.ibm.com/think/insights/artificial-intelligence-advantages-disadvantages)

****__Q. What will we produce?__****

- Powerpoint presentation
- Practical demonstration (Generative AI tool)
- Interative activity (Kahoot maybe?)